

CELL AND MOLECULAR BIOLOGY LABORATORY

From Gene Mutation to Disease

Investigating How DNA Sequence Changes Affect Protein Products and Human Phenotypes

Background:

Many human diseases and biological traits are associated with changes in DNA sequence. A mutation may involve one nucleotide or many nucleotides. It may change an amino acid, introduce a premature stop codon, shift the reading frame, remove amino acids, add amino acids, or affect gene function in another way.

In this laboratory, you will investigate one human disease or medically relevant phenotype and one associated gene. You will study the normal function of the gene, obtain its normal coding sequence (CDS), and translate it to obtain the predicted normal protein sequence, manually reproduce a documented mutation, translate the mutant sequence, and compare the normal and mutant proteins.

Your main goal is to explain the following relationship: DNA mutation → change in coding sequence → change in protein → change in molecular or cellular function → disease or phenotype.

Learning Objectives

1. Investigate the molecular basis of a human genetic disease or phenotype.
2. Retrieve an appropriate human reference CDS and protein sequence.
3. Explain the normal function of a selected gene and protein.
4. Identify a documented genetic variant associated with a disease or phenotype.
5. Manually reproduce a mutation in a nucleotide sequence.
6. Translate normal and mutant coding sequences.
7. Compare normal and mutant protein sequences.
8. Identify mutation types such as missense, nonsense, frameshift, in-frame insertion/deletion, and repeat expansion.
9. Predict how a mutation may affect protein function.
10. Relate molecular changes to cellular and organismal phenotypes.
11. Distinguish computational prediction from experimental evidence.
12. Document the laboratory workflow using Galaxy and GitHub.

Disease: SCN9A Congenital Insensitivity to Pain

Disease Background:

Channelopathy-associated Congenital Insensitivity to Pain (CIP) is an exceptionally rare genetic disorder characterized by a complete, lifelong inability to perceive physical pain from birth. Affected individuals cannot feel noxious stimuli resulting from severe trauma, deep burns, bone fractures, or internal inflammatory conditions such as appendicitis. Additionally, most patients present with anosmia (a loss of the sense of smell) because the underlying channel defect also affects olfactory neurons. Due to the lack of protective pain sensation, young children frequently experience unintentional self-inflicted injuries, including severe biting of the tongue, lips, and cheeks during teething. Over time, undetected micro-trauma often leads to serious orthopedic complications such as joint destruction (Charcot joints), chronic osteomyelitis, and poorly healed fractures. Despite these severe sensory deficits, tactile perception—including touch, pressure, vibration, and temperature differentiation—remains functional, and individuals typically demonstrate normal cognitive development, motor control, and autonomic function (such as normal sweating and heart rate regulation).

Gene and Normal Protein Function:

Item	Answer
Official gene symbol	SCN9A
Official full gene name	sodium voltage-gated channel alpha subunit 9
Organism	Organism Homo sapiens (Human)
NCBI Gene ID	6335
Gene type (for example, protein-coding)	protein-coding
NCBI Gene record URL	https://www.ncbi.nlm.nih.gov/datasets/gene/6335/

Part II. Research the Disease

Before changing any sequence, understand the biology of your disease and gene.

A. Disease

13. What is the disease or phenotype?

- Congenital Insensitivity to Pain (CIP) (specifically *Channelopathy-associated Congenital Insensitivity to Pain*). It is an extremely rare genetic condition characterized by the complete inability to perceive physical pain from birth.

14. What are its major clinical characteristics?

- Inability to feel physical pain: Affected individuals do not feel pain from injuries, burns, fractures, or internal conditions like appendicitis. Loss of smell (Anosmia): Often present as a co-occurring trait due to the role of the same sodium channels in olfactory neuron. Unintentional self-injury: Frequent tongue, lip, or cheek biting during early childhood; repeated undetected burns, bruises, cuts, and bone fractures. Joint and bone complications: Unnoticed micro-trauma leading to joint destruction (Charcot joints), chronic osteomyelitis, and delayed healing of unrecognized fractures. Preserved tactile senses: Touch, vibration, pressure, and proprioception (body positioning) remain intact. Normal intelligence and motor function: Cognitive development, muscle strength, and autonomic nervous system function (sweating, heart rate, and temperature regulation) are typically normal (unlike conditions such as HSN/HSAN type IV).

15. Which cells, tissues, or organs are mainly affected?

- Cells: Nociceptors (first-order sensory neurons located in the dorsal root ganglia) and olfactory sensory neurons in the nasal cavity. Tissues & Organs: The peripheral nervous system (PNS)—specifically the sensory nerve fibers responsible for carrying pain signals to the central nervous system. Secondary damage occurs in the skin, bones, joints, and oral mucosa due to unperceived trauma.

16. What is its genetic basis?

- Driven by homozygous or compound heterozygous loss-of-function mutations in the SCN9A gene (located on chromosome 2q24.3). The SCN9A gene encodes the Na1.7 voltage-gated sodium channel alpha subunit. These mutations cause a complete loss of function in the Na1.7 channel, preventing the generation and propagation of electrical action potentials in response to painful stimuli.

17. What is its inheritance pattern, if applicable?

- Autosomal Recessive: An affected individual inherits two mutated copies of the SCN9A gene (one from each heterozygous parent, who are asymptomatic carriers).

B. Gene and Normal Protein

18. What is the official gene symbol?

- SCN9A

19. On which human chromosome is the gene located?

- Chromosome 2 (specifically at cytogenetic band 2q24.3).

20. What does the gene normally encode?

- It encodes the alpha Nav1.7 of a voltage-gated sodium channel protein. This subunit forms the functional, pore-forming core of the channel.

21. What is the normal biological function of the protein?

- Initiating & Amplifying Pain Signals: Acts as a threshold channel that opens in response to small depolarizations at nerve endings, amplifying subthreshold depolarizing inputs to generate an action potential. Sodium Ion Transport: Selectively conducts positively charged sodium ions into sensory neurons, driving rapid cellular depolarization necessary for electrical signal conduction.

22. Where in the cell is the protein normally found?

- Plasma Membrane: Primarily located within the cell membrane of primary sensory neurons (dorsal root ganglion nociceptors) and olfactory sensory neurons. Axonal Terminals & Fibers: Concentrated along peripheral unmyelinated nerve endings (C-fibers) and small myelinated nerve fibers (delta-fibers) where pain stimuli are first detected.

23. In what biological pathway or cellular process does it participate?

- **Pathway:** Sensory perception of pain (Nociception) and olfactory signaling. **Cellular Process:** Voltage-gated sodium channel activity, action potential generation and propagation, and neuronal signal transduction within the peripheral nervous system.

C. Documented Mutation

Gene	SCN9A
Reference Transcript	NM_002977.3
Exact Variation Notation	c.2674C>T
Nucleotide Change	Cytosine to Thymine transition at position 2674 of the coding sequence (C to T)
Predicted Protein Change	p.Arg892Ter
Mutation Type	Nonsense mutation
ClinVar Accession (when available)	VCV000007255
Clinical Interpretation	Pathogenic (Loss-of-Function causing Congenital Insensitivity to Pain)
Scientific References Supporting the Association	Cox, J. J., et al. (2006). An SCN9A channelopathy causes congenital inability to experience pain. <i>Nature</i> , 444(7121), 894–898.

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FASTA

Homo sapiens sodium voltage-gated channel alpha subunit 9 (SCN9A), transcript variant 1, mRNA

NCBI Reference Sequence: NM_002977.4

[GenBank](#) [Graphics](#)

>NM_002977.4 Homo sapiens sodium voltage-gated channel alpha subunit 9 (SCN9A), transcript variant 1, mRNA

AGTCTGCTTGACAGCGGTCGACGCGCTCCAGCGCGGCTGTCGGCTTTCCAATTCGCCAGCTCGGCTG
AGGCTGGGCTAGCCTGGGTGCCAGTGGCTGCTAGCGGCAAGCGCTCCCCTGAGCAACAGGAGCCAGAGAA
AAAGAAGCAGCCCTGAGAGAGCGCCGGGGAAGGAGAGCGCCGCGCCCTCTCCTGGAGCCAGATTCTGCAG
GTGCAC TGGGTGGGATGATCGGCGGCTAGGTTGCAAGCCTCTTATGTGAGGAGCTGAAGAGGAATTAA
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GACATTCTCCCGCATGGTGTGAGAGCCCTGGAGGACTTGGACCCCTACTATGACAGACAAAAGACTT
TCATAGTATTGAACAAAGGGAACAATCTCCGTTTCAATGCCACACCTGCTTTATATATGCTTTCTCC

Analyze this sequence

[Run BLAST](#)

[Pick Primers](#)

[Show in Genome Data Viewer](#)

Reference sequence information

RefSeq alternative splicing
See 14 reference mRNA sequence splice variants for the SCN9A gene.

Figure 1. Obtain the Normal Reference Sequence

Galaxy

Regular Composite Collection Rule-based

Please wait...1 out of 1 remaining...

SCN9A_c.2674CT_C 6.1 KB Auto-detect unspecified (?) 100%

Type (set all): Auto-detect Reference (set all): unspecified (?)

New upload

Choose local file Choose from repository Paste/Fetch data

Figure 2. . Import the Normal Sequence into Galaxy

Part V. Establish the Wild-Type Control

Item	Predicted Protein	Reference Protein
Record the CDS length	5,934 bp	5,934 bp
Predicted protein length	1,977 aa	1,977 aa
Start codon	ATG	ATG
Stop codon	TAG	TAG
Reading frame	Frame +1	Frame +1
First 10aminoacids	MAMLPPPGPQ	MAMLPPPGPQ
Last 10 amino acids	EDKEKDSKESKK	EDKEKDSKESKK

Part VI. Formulate a Mutation Hypothesis

Item	Predictions
Mutation and exact nucleotide change	c.2674C>T (Cytosine replaced by Thymine at position 2674 in the CDS)
Number of nucleotide(s) affected	1 nucleotide
Predicted mutation type	Nonsense mutation
Predicted effect on the reading frame	No shift
Predicted effect on protein length	Truncated protein
Predicted effect on protein function	Complete loss of function

Part VII.

Item	Answers
Mutant CDS length	5,934 bp
Mutant protein length	1,977 aa
Reading frame	Frame +1
Location of the first amino- acid difference	Position 892
Presence or absence of a premature stop codon	Absent
Approximate number of amino acids	1 amino acid

Part IX. Compare WT and Mutant Proteins

Compare the WT protein and mutant protein using an appropriate sequence alignment method when needed.

24. At what amino-acid position do the sequences first differ?

- Amino acid position 892.

25. Is only one amino acid affected?

- Yes. Only codon 892 is altered (converting ATC to ATG, which changes Isoleucine to Methionine: p.I892M).

26. Are multiple downstream amino acids changed?

- No. All downstream amino acids remain identical to the wild-type protein sequence.

27. Was an amino acid deleted or inserted?

- No. It is a single-base substitution, so no amino acids were inserted or deleted.

28. Was a premature stop codon produced?

- No. The substitution results in a functional codon coding for Methionine rather than a stop codon (TAA, TAG, or TGA).

29. Did the reading frame change?

- No. Because there were no insertions or deletions (indels) of nucleotides, the open reading frame remains unchanged (Frame +1).

30. Did the protein length change?

- No. Both the wild-type and mutant proteins are 1,977 amino acids long.

31. Is the mutation missense, nonsense, frameshift, in-frame deletion/insertion, repeat expansion, or another type?

- Missense mutation specifically a single nucleotide substitution resulting in an amino acid change.

Part X. Explain the Molecular Consequence

A single-nucleotide point mutation in the *SCN9A* gene (ATC - ATG at codon 892) alters the DNA sequence without shifting the reading frame, resulting in a missense substitution of isoleucine to methionine (p.I892M) in the Nav 1.7 sodium channel. Located in the DII/S5 transmembrane segment, this structural change disrupts channel gating and produces a gain-of-function effect by hyperpolarizing the activation threshold, allowing the channel to open more easily. At the cellular level, this altered kinetics causes dorsal root ganglion (DRG) sensory neurons to become hyperexcitable, firing repetitive action potentials in response to minimal stimuli. Ultimately, this sustained propagation of nociceptive signals manifests clinically as Inherited Erythromelalgia (IEM), causing severe, episodic burning pain, redness, and heat sensation in the extremities.

Part XI. Second Experiment: Create Your Own Mutation

Target Position	Nucleotide 2674
Wild-type	CGA (Arginine)
Mutant	TGA (Substitution of C to T)

Changing a single nucleotide at position 2674 from cytosine to thymine (CGA - TGA) converts the codon for arginine into a premature stop codon, creating a **nonsense mutation**. Because this single-base swap does not add or remove any nucleotides, the reading frame remains unchanged. However, during translation, the ribosome encounters this premature stop signal at position 892 and halts protein synthesis early. This cuts off over 1,000 downstream amino acids, producing a severely truncated, non-functional protein.

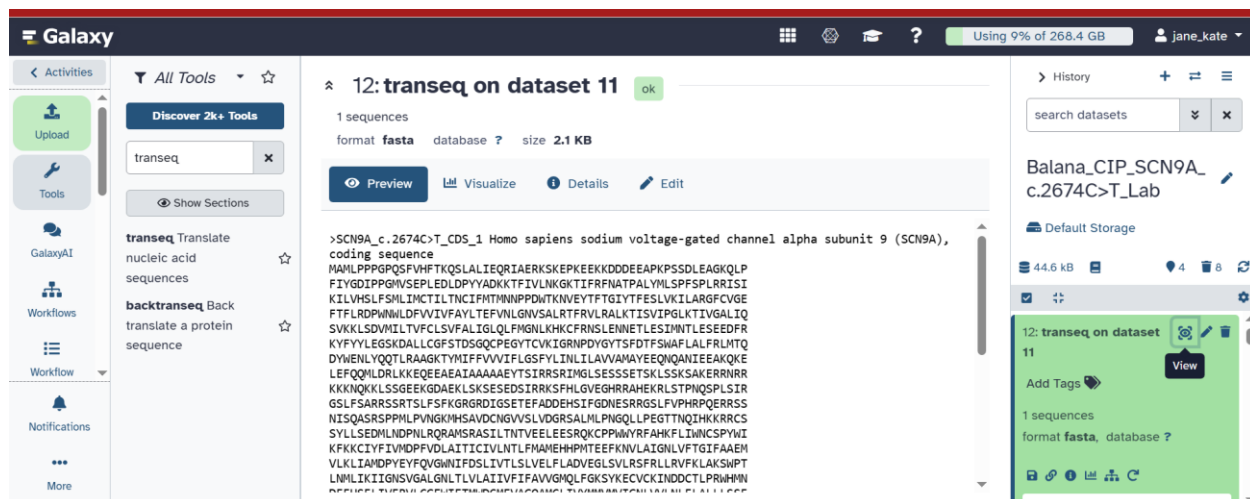


Figure 3. Create the Mutant Sequence and Translate the Mutant Sequence

Part XII. Compare the Documented and Artificial Mutations

Item	Wildtype	Documented Mutation
CDS length	5,934 bp	5,934 bp
Protein length	1,977 aa	1,977 aa
Mutation type	None	Single-nucleotide substitution
Reading frame changed	No	No
Premature stop codon appeared	No	No
Amino acids affected	None	1 amino acid (p.I892M)
Expected functional	Normal channel gating and function	Gain of function causing neuronal hyper excitability and Inherited Erythromelalgia (IEM)

Part XIII. Interpretation Questions:

32. Why does the exact location of a mutation matter?

- The location determines which functional region or domain of the protein is altered. A mutation in a critical area—such as an active site, a binding domain, or a pore-forming trans membrane helix—can severely disrupt structural stability or activity, whereas a mutation in a flexible loop or non-essential region may have little to no functional impact.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

- Genetic code is read in non-overlapping triplets (codons). Deleting three nucleotides removes exactly one codon, preserving the downstream reading frame (in-frame deletion). Deleting one or two nucleotides shifts the codon grouping for all downstream bases (frameshift mutation), altering every subsequent amino acid and usually introducing a premature stop codon.

34. Does every mutation change the amino-acid sequence? Explain.

- No. Because the genetic code is redundant (degenerate), multiple codons can code for the same amino acid. A substitution that changes a codon to another codon for the same amino acid is a synonymous (silent) mutation, leaving the protein sequence unchanged.

35. Does every amino-acid substitution destroy protein function? Explain.

- No. A conservative substitution (replacing an amino acid with one of similar chemical properties, such as charge or hydrophobicity) in a non-critical domain may have no measurable effect. Furthermore, some substitutions only subtly alter protein kinetics or activity without destroying function entirely.

36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?

- Deleting one nucleotide shifts the reading frame of every subsequent triplet downstream. Because every single codon following the deletion site is now read differently by the ribosome, all downstream amino acids are completely changed until a stop codon is encountered.

37. Why might a premature stop codon produce a nonfunctional protein?

- It terminates translation early, leaving the protein truncated and missing crucial downstream functional domains needed for proper folding, catalysis, or channel gating. Additionally, cells often destroy such truncated transcripts via **nonsense-mediated mRNA decay (NMD)**, preventing protein synthesis altogether.

38. Could a mutation affect protein function without greatly changing protein length?

- Yes. A single missense mutation alters only one amino acid without changing protein length, yet it can drastically alter channel kinetics, enzymatic activity, or protein stability (as seen with the gain-of-function p.I892M substitution in NaV1.7).

39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.

- Yes. A non-coding mutation in a promoter or enhancer region can disrupt transcription factor binding, leading to extreme under- or over-expression of a normal protein. Alternatively, a

mutation in a splice donor/acceptor site can cause aberrant pre-mRNA splicing (e.g., exon skipping or intron retention) without directly altering a coding codon in the original transcript.

40. What evidence from your analysis supports the proposed molecular mechanism of your disease?

- Computational translation and sequence alignment confirmed that the single-base substitution causes a single missense change (p.I892M) at codon 892 in the DII/S5 domain of NaV1.7, leaving the full length (1,977 aa) and open reading frame intact without causing a frameshift or premature truncation.

41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?

- Supported directly by computational results: Sequence alignment, identifying the mutation type (missense), determining codon/reading-frame stability, locating the affected amino acid position (892), and calculating predicted protein length. Moreover, it requires experimental studies: Electrophysiological properties (voltage-gated activation thresholds, ramp currents), cellular hyper excitability in DRG neurons, physical protein tertiary structure, and actual clinical phenotype/disease expression (Inherited Erythromelalgia symptoms).

REFERENCES:

Dib-Hajj, S. D., Yang, Y., Black, J. A., & Waxman, S. G. (2013). The NaV1.7 sodium channel: From channelopathy to pain therapeutics. *Nature Reviews Neuroscience*, 14(1), 49–62.
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GALAXY HISTORY LINK:

https://usegalaxy.org/u/jane_kate/h/copy-of-copy-of-balana-cip-sc9a-c-2674c-t-lab